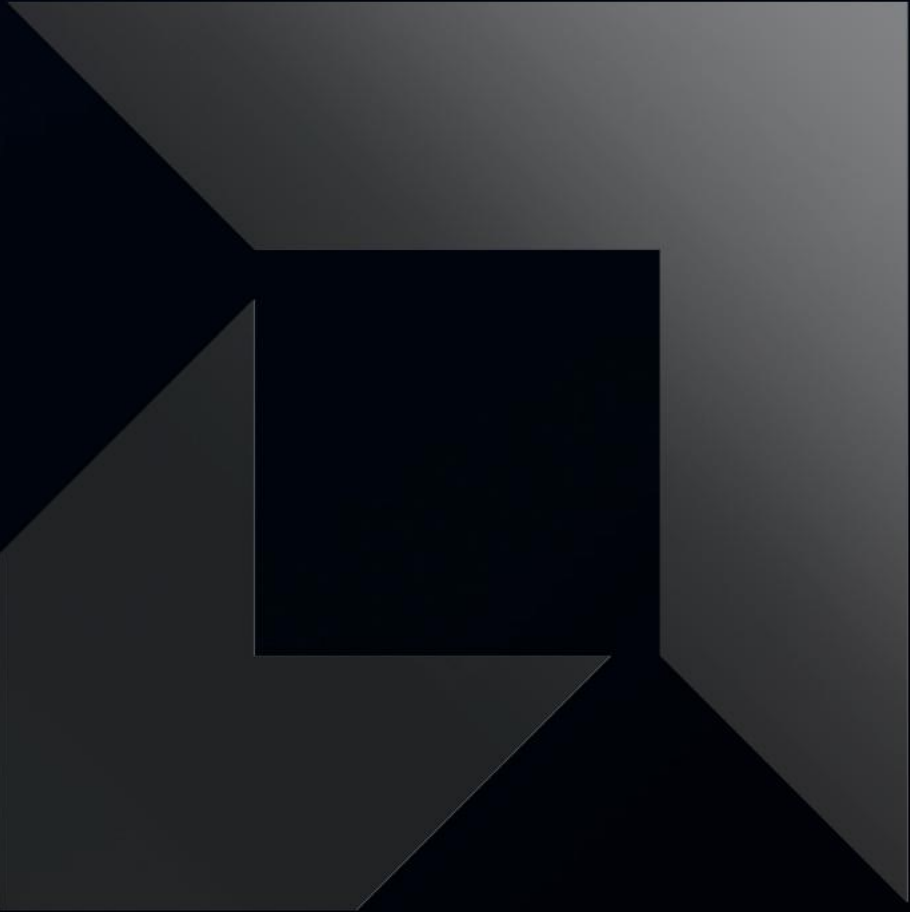




# Reference Cluster Design Architecture MI300X / MI325X

128 GPU to 1024 GPU

Version 4.1 – February 2026

# Cluster - 128 GPU to 1024 GPU (16 to 128 Nodes) \*

| Cluster Size | 128 to 1024 GPU                                   |
|--------------|---------------------------------------------------|
| Platforms    | Dell XE9680<br>Lenovo SR685a V3<br>SMCI AS-8125GS |
| OS           | Ubuntu 22.04 (or above)                           |
| Linux Kernel | 5.15 – 6.80                                       |
| ROCm         | 6.3.3 (or above)                                  |

\*Genericized BoM – removing device specifics including dependencies

| Storage Type           |                                |
|------------------------|--------------------------------|
| Local Storage          | 1.6 TB (or greater)            |
| Utility Storage        | Pure, Vast, RYO                |
| Bulk Storage           | Pure, Vast, WekaIO             |
| Scratch Storage        | Vast, DDN, WekaIO, Hammerspace |
| Archive/Object Storage | S3 Compatible                  |

# Network - 128 GPU to 1024 GPU (128 to 1024 Nodes)\*

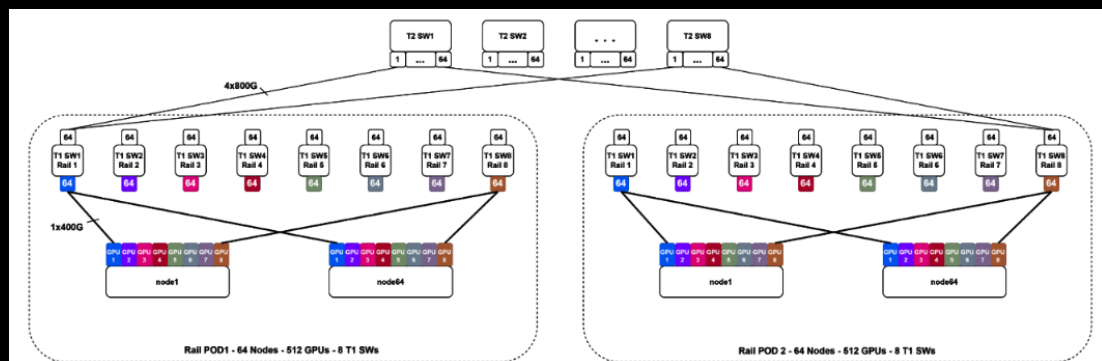
| Backside Network Topology | 2 Tier Rail Optimized / Fat Tree                            |
|---------------------------|-------------------------------------------------------------|
| NIC                       | Pollara 400<br>BCM957608 (Thor2)                            |
| Switch                    | Arista, Dell, Juniper, Cisco<br>(TH 4/5, Scheduled fabrics) |
| Network OS                | SONiC, Junos, EOS, IOS,<br>DriveNets                        |
| Subscription Ratio        | 1:1.16=16% Undersubscribed<br>(AMD recommendation)          |
| Optics                    | Vendor ACL/HCL transceivers<br>or Direct Attach Copper      |
| Fabric                    | RoCEv2 Ethernet                                             |

| Frontside Network Segment         | Adapter Recommended                    |
|-----------------------------------|----------------------------------------|
| All-in One Network                | Ethernet 100GbE 2-port QSFP28 Adapter  |
| Storage Network (Optional)        | Ethernet 100GbE 2-port QSFP28 Adapter  |
| Virtualization Network (Optional) | Ethernet 100GbE 2-port QSFP28 Adapter  |
| Host In-Band                      | Ethernet 10/25GbE 4-Port SFP28 Adapter |
| BMC OOB Mgt                       | 1G Copper                              |

\* Genericized BoM – removing device specifics including dependencies

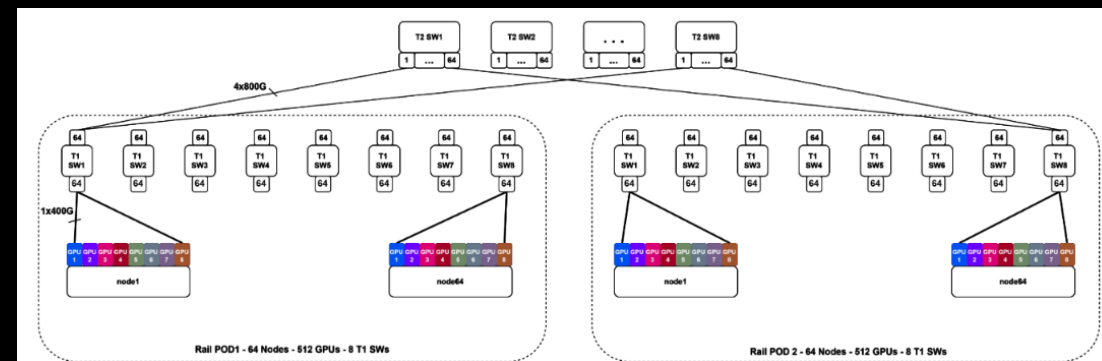
# 128 GPU to 1024 GPU – General Network Layouts

## Rail Optimized (1:1 example)



- Rail design - 2 pods
- 64 Nodes per pod – 512 GPUs
- Sizing of network can be done through lower/increase spine layer and adjusting pod size.
- Lower latency for AI workloads – optimized CPU communication
- Improved performance for large scale models

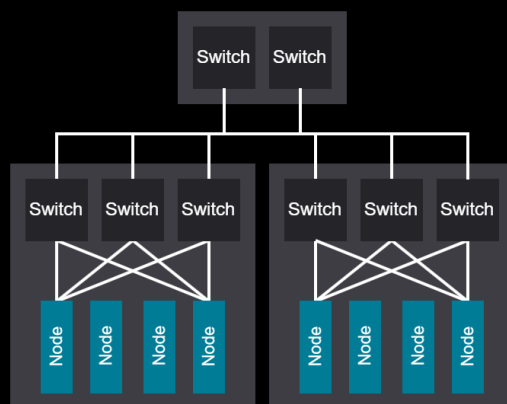
## Tree Topology (1:1 example)



- Tree design – 2 pods
- 64 Nodes per pod – 512 GPUs
- Sizing of network can be done through lower/increase spine layer and adjusting pod size.
- Non blocking bandwidth and scalability
- Load balancing and fault tolerance
- Diverse workloads

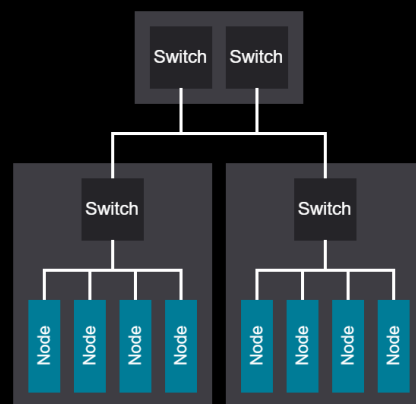
# Cluster Network

# Basic Network Topologies



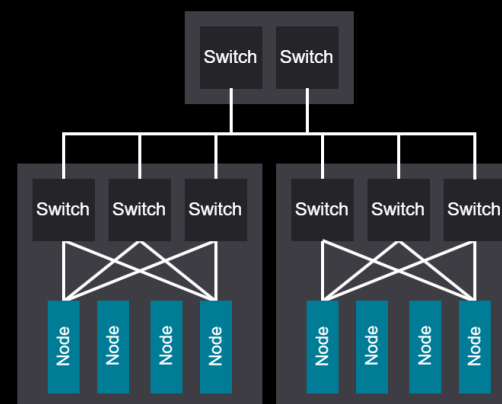
## 2-Tier Rail Network

- Enables large scalable unit sizes for large jobs or replica sizes
- Efficient for workloads favoring ring-based collectives
- Higher infrastructure costs



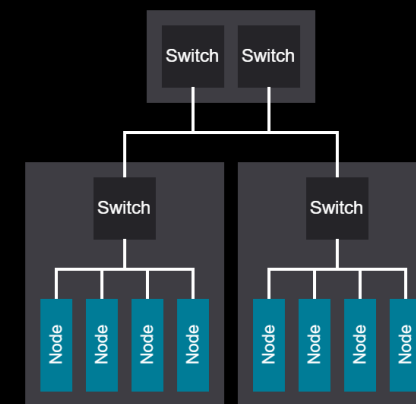
## 2-Tier Tree Network

- Efficient for small workloads or replicas
- Easy to add capacity with proper planning
- Potentially allows for lower infrastructure costs
- Limited blast radius compared to rail networks



## 3-Tier Rail TH5/J3 Network

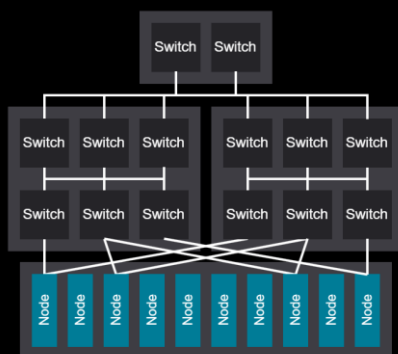
- Spine switches replaced with a 2-tier Jericho3-AI/Ramon3 fabric for increased maximum cluster size
- Deep buffers and scheduled fabric aid greatly with congestion issues in large clusters at a small latency penalty



## 3-Tier Tree TH5/J3 Network

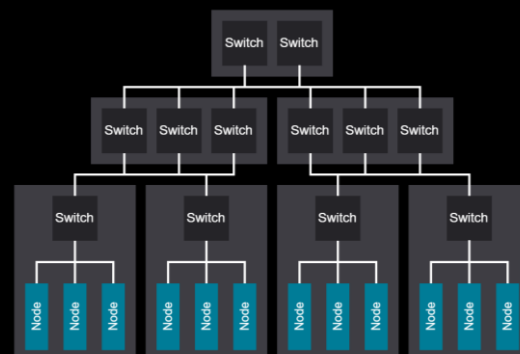
- Same benefits from switch to scheduled spine fabric as with rail, but retains the primary characteristics of tree networks

# Basic Network Topologies Continued



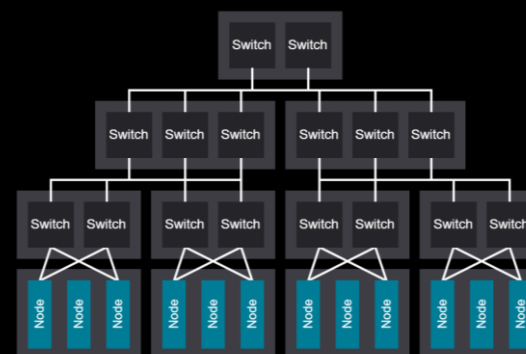
## 3-Tier Rail Optimized Network

- Allows for massive scalable unit sizes
- Best ring-based collective performance at scale; at the cost of poor any-any performance



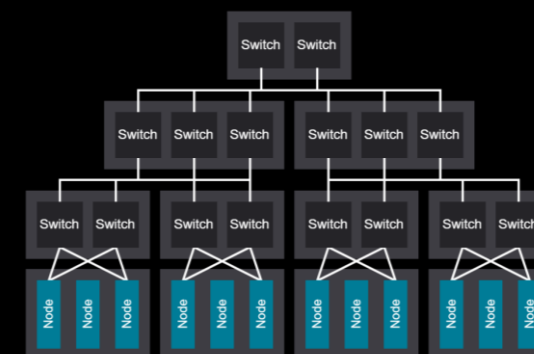
## 3-Tier Tree Network

- Allows for massive cluster sizes
- Best any-any performance at scale
- Suitable for a “campus style” deployment



## 3-Tier Hybrid Rail Network

- Allows for massive cluster sizes with large scalable units
- Favors ring-based collectives, but does not sacrifice significant any-any performance on large jobs
- Suitable for a “campus-style” deployment



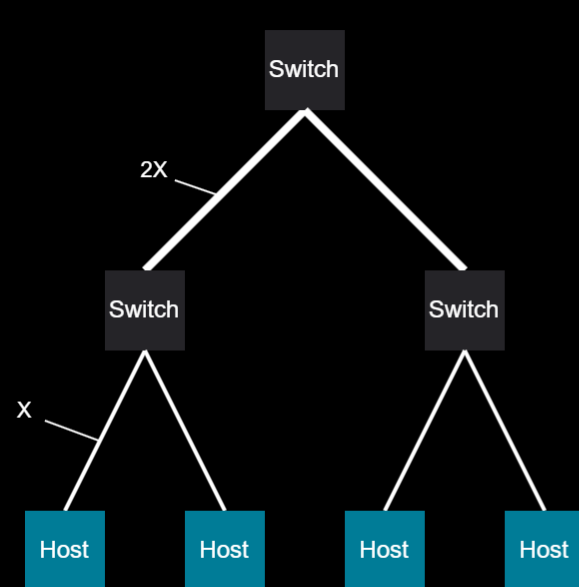
## 3-Tier Fully Scheduled Rail Network

- Medium sized scalable units
- Excellent congestion performance due to deep buffers and scheduled fabric
- Technical limitations limit cluster size to ~32K GPUs in recommended configuration

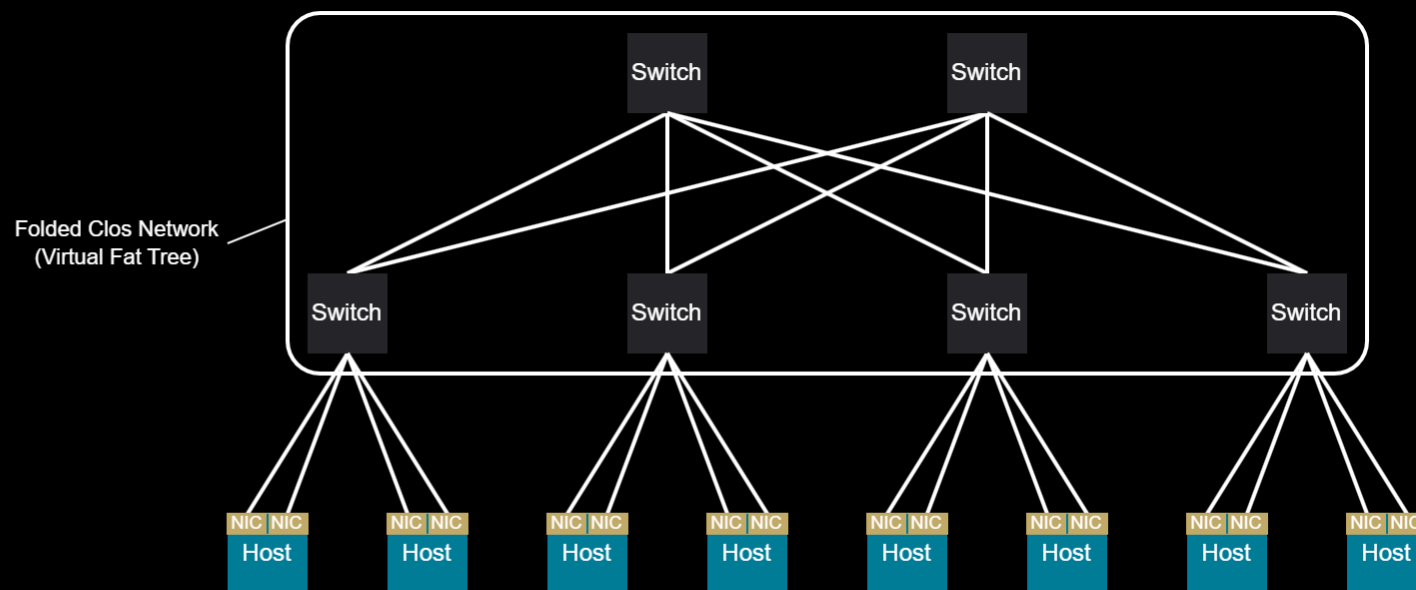
# Tree and Rail



# Fat Tree Terminology In Cluster Design



Fat Tree Topology

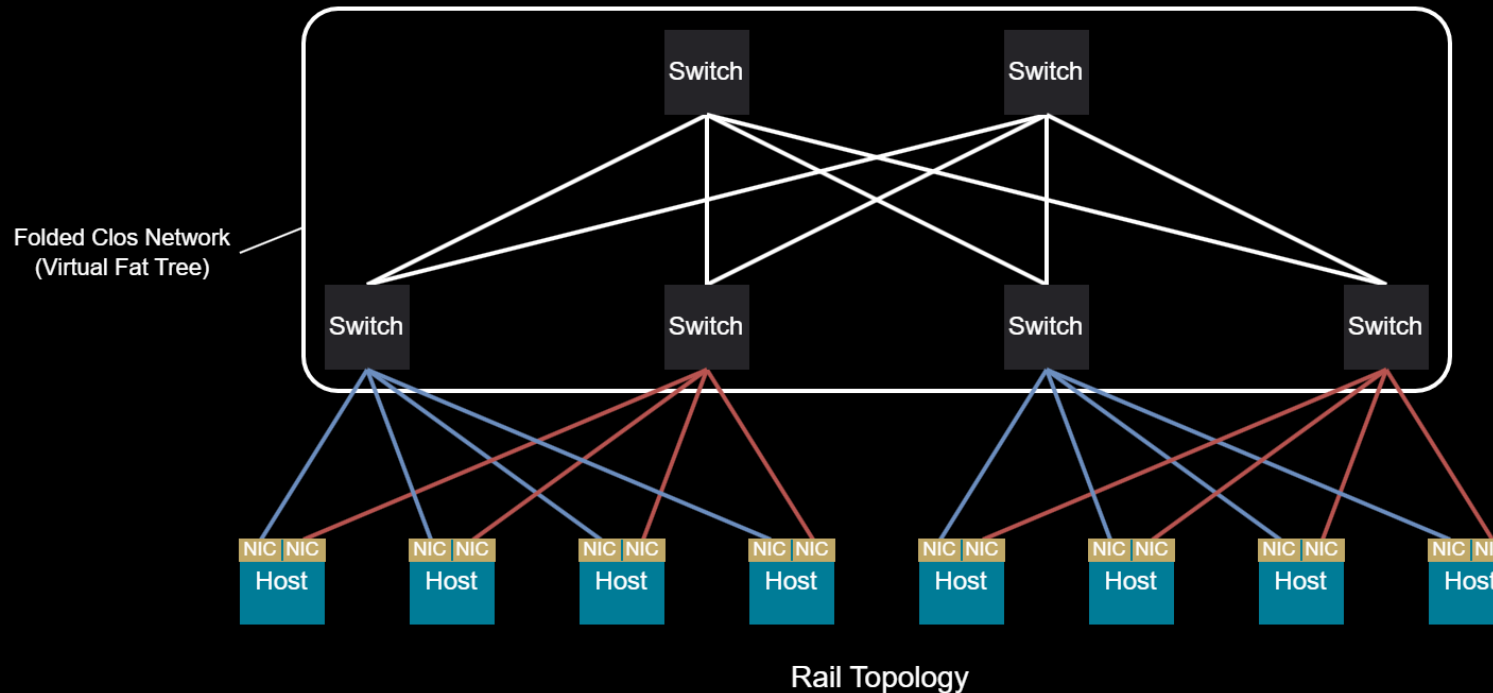


"Fat Tree" Topology

The canonical fat tree topology is a network concept where a switch's connection to upstream peers has at least parity bandwidth with the total aggregate bandwidth of its downstream connections. This causes links between switches to become "fatter" as they get closer to the core.

The "fat tree" topology for AI/ML clusters instead refers to how a host is connected to its upstream switches; in this case all host NICs terminate on the same switch. It can also be considered 1-rail network. The network itself is generally a 3-stage or 5-stage folded Clos network due the fixed radix of network switches.

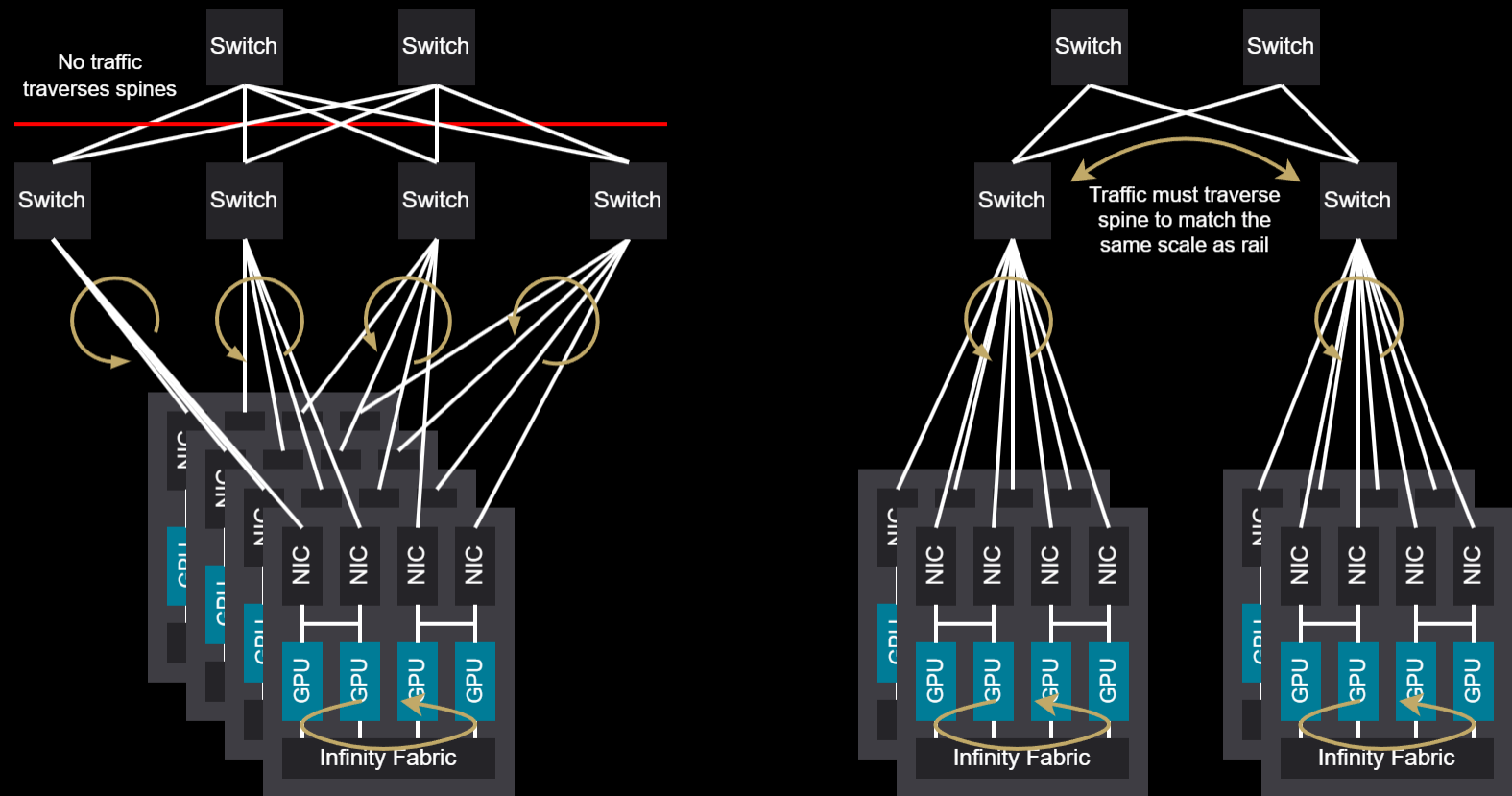
# Rail Networks In Cluster Design



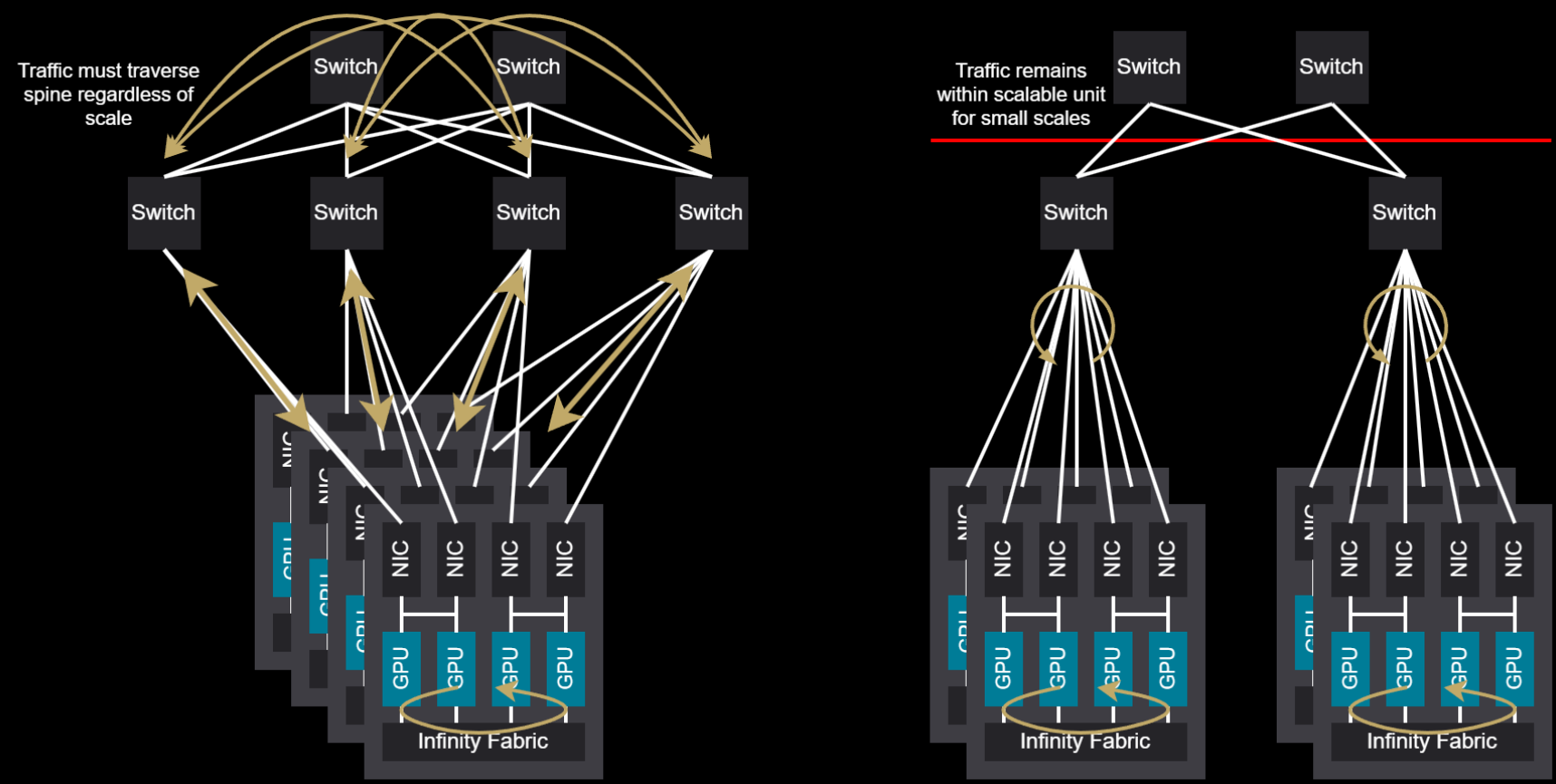
Rail networks leverage the same folded Clos network as tree networks, but host connections are instead aggregated onto switches based on NIC rank. These shared ranks are referred to as rails and allow the network to provide preferential latency for connections which share the same rail. The downside to this design is any traffic which needs to cross rails/ranks must traverse either the network spine layer, or Infinity Fabric (PXN).

The above example shows an example 2-rail network, with the rails colored blue and red to differentiate them.

# Rail Enables Larger Single Hop Ring Domains

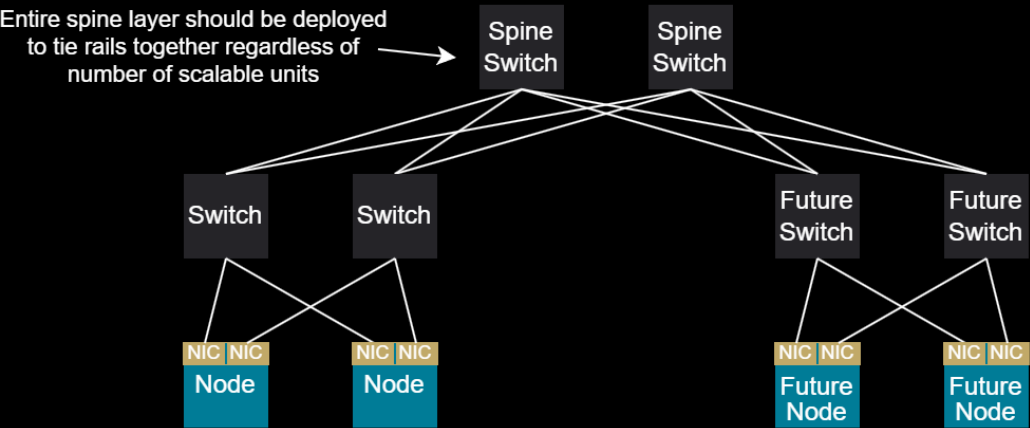
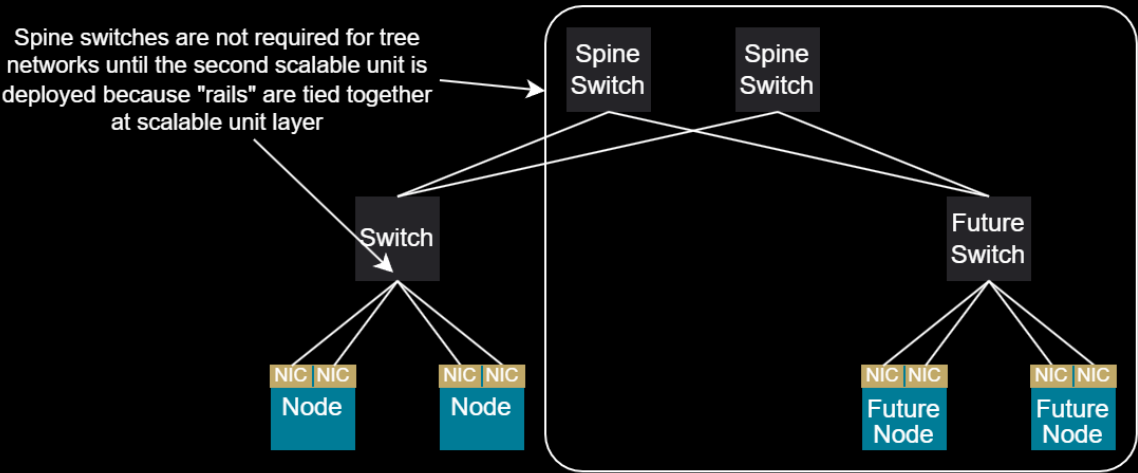


# Tree Handles Cross-Rank Traffic Better

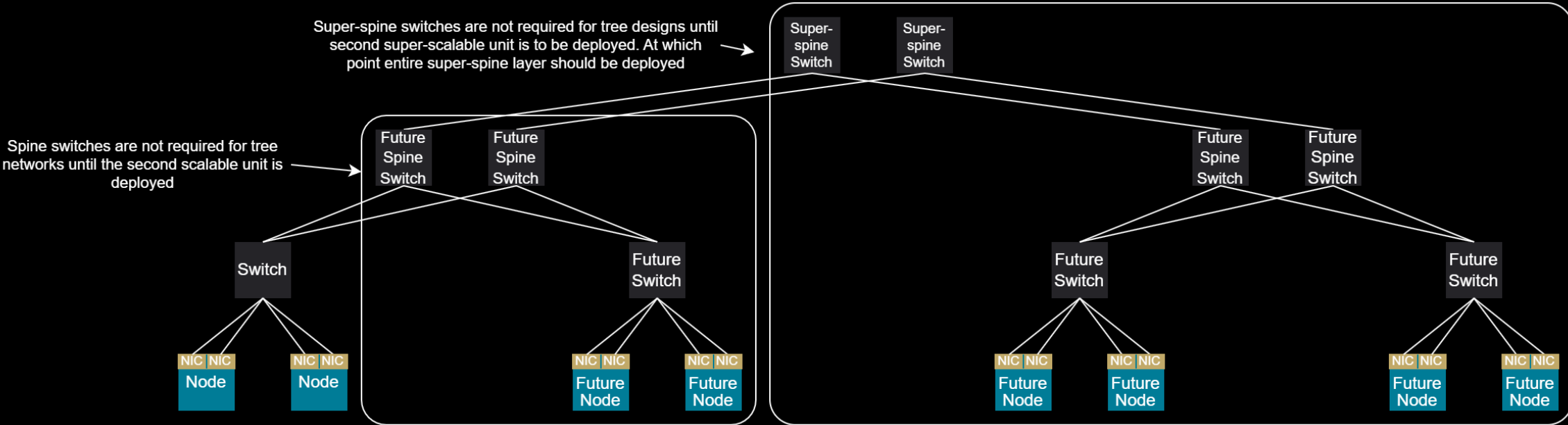


# Scaling Networks

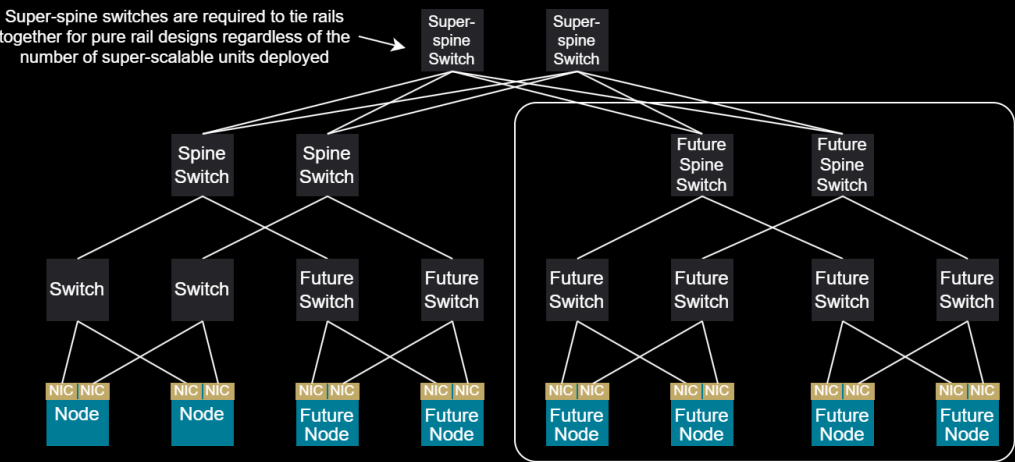
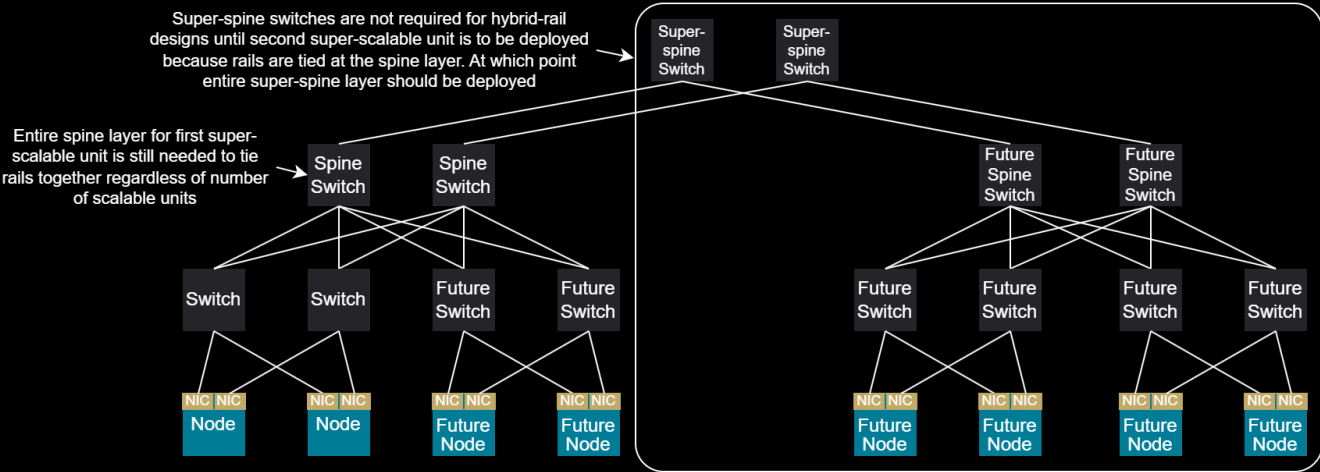
# Cluster Backend Deployment Strategy for 2-tier Networks



# Cluster Backend Deployment Strategy for 3-tier Tree Networks



# Cluster Backend Deployment Strategy for 3-tier Rail Networks





# Network Subscription

Subscription is the relationship between what is provided by the upstream network and what is required by the downstream network in demand side.

It is typically represented as a ratio:

$$\text{Downstream Demand} : \text{Upstream Capacity}$$

1:1 subscribed network would be equal downstream capacity to upstream capacity.

Example: 1:1.16 = (1 downstream demand : 1.16 upstream capacity. In this example there is .16 more upstream capacity)

Or as a percentage:

$$\text{Subscription Rate} = \frac{\text{Downstream Demand}}{\text{Upstream Capacity}}$$

80% subscription ratio could be referred to as “20% undersubscribed”, or a 120% subscription ratio could be referred to as “20% oversubscribed”.

# Network Design Examples

# Network Topology Design Notations:

## Design note:

Designs included are based on scheduled fabrics switch type (Arista, Ciena, Nokia) or 51.2T switch type (Arista, Cisco, Dell, Juniper)

Vendors/switch models vary for port count and features – please consult desired vendor port count directly to confirm.

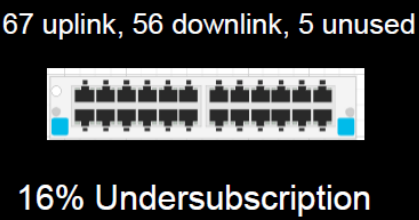
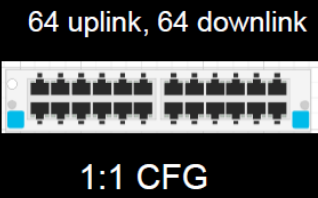
## Scalable Units/PODs note:

Diagrams presented are designed around a Scalable Unit or POD – which can determine overall network end to end latency and AI use cases.

Certain ML/AI workloads may require change of scalable unit size. Please consult with AMD Architecture as required.

# Network Layout - 128 GPU Cluster 16 Nodes – Single Switch

Example 128 port switch



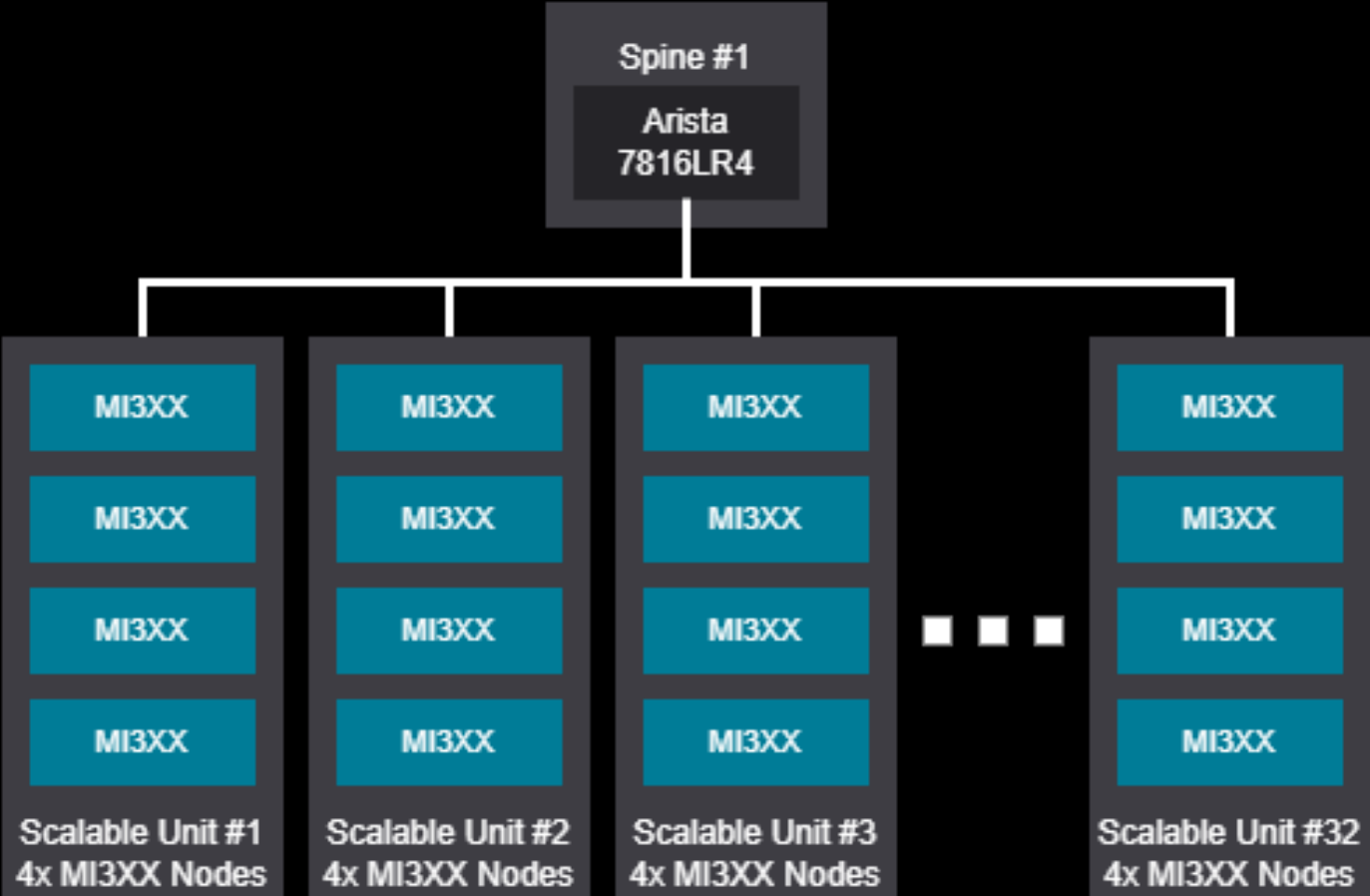
## Fat Tree / Rail

| Tier   | Leaf switch | Spine switch | Superspine switch |
|--------|-------------|--------------|-------------------|
| 3 tier | 16%         | 1-1          | All               |

# Network Diagram - 128 GPU to 1024GPU (16 to 128 Nodes)

## Tree Design

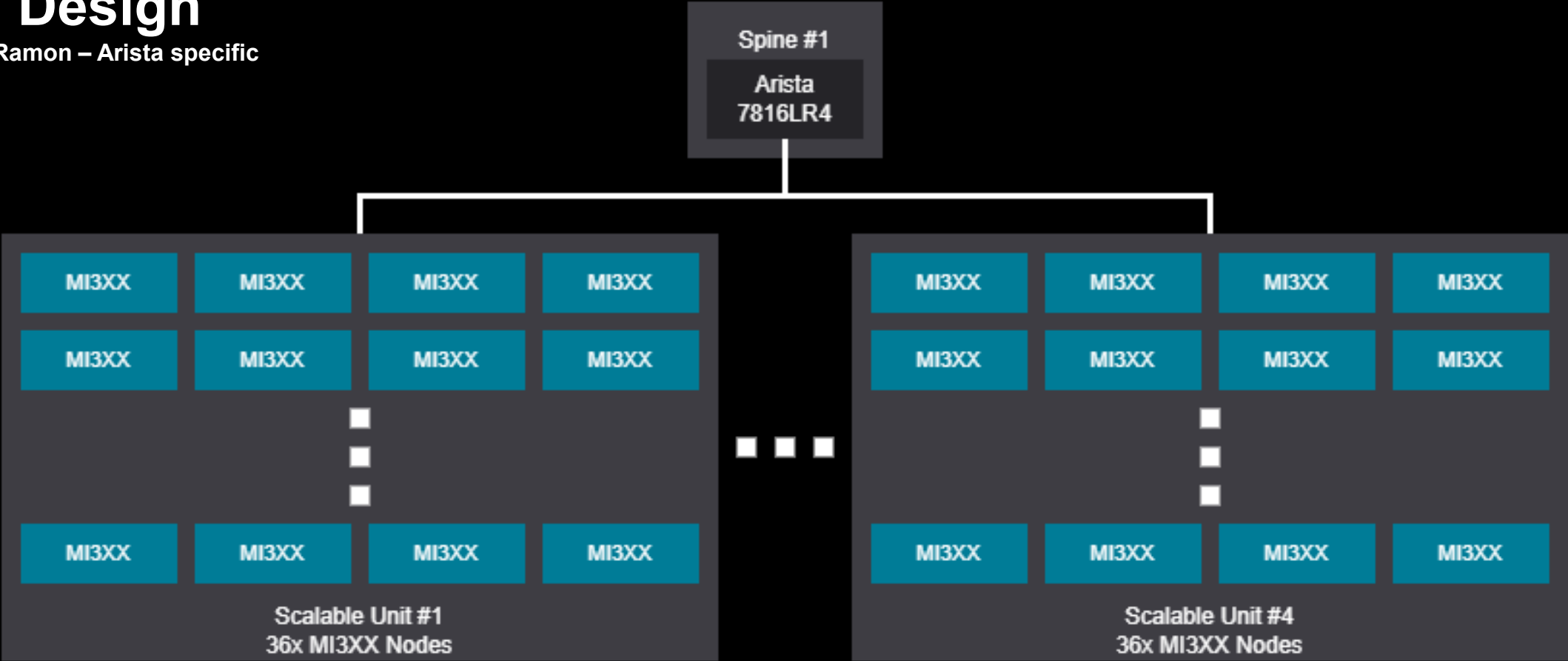
Jericho/Ramon – Arista specific



# Network Diagram - 128 GPU to 1152GPU (16 to 144 Nodes)

## Rail Design

Jericho/Ramon – Arista specific



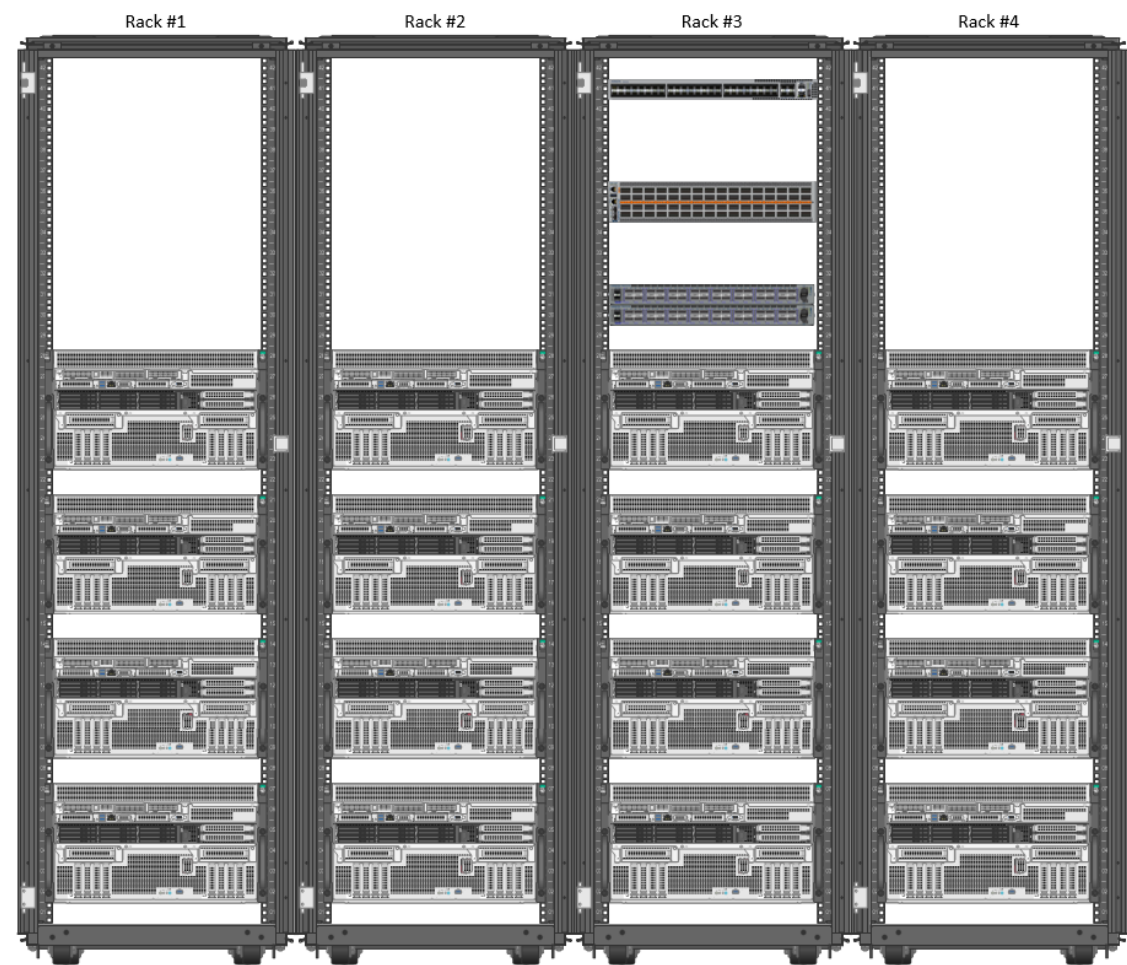


# Appendix 1 – Vendor Specific Designs



# HPE – Arista Design – Sample Racks (POD)

# Sample Rack Elevation - 128 GPU (16 Nodes) - Single Switch



**Rack #1, #2, #4 – GPU Racks**  
4 x HPE XD685 MI325 GPU Nodes  
**Rack Power Estimate: 52 KW**

**Rack #3 – GPU + Network Rack**  
4 x HPE XD685 MI325 GPU Nodes  
1 x Arista TH5 800G 64p Switch – L1 BE Net  
2 x Arista 32p 100G Switch – FE Net  
1 x Arista 48p 1G Switch – OOB MGMT  
**Rack Power Estimate: 55 KW**

**Hardware Power Estimate**  
MI325 GPU Node – 13KW  
Arista TH5 800G 64p Switch – 2250W  
Arista 100G 632p Switch – 350W  
Arista 1G 48p Switch – 95W

**Total Solution Power Estimate: 211 KW**

|                        |               |            |      |
|------------------------|---------------|------------|------|
| 128 GPU MI325X Cluster |               |            |      |
|                        | INTL          | DATE       | AMD  |
| DRAWN BY:              | BGM           | 03/01/2025 |      |
| APPROVED BY:           | ---           | --/--/---- |      |
| SFDC ID:               | OPPORTUNITY # | SHEET      | OF N |

# Sample Rack Elevation - 512GPU (64 Node) POD- Rail-Optimized



# Sample Rack Elevation - 1024GPU (128 Node) POD- Rail-Optimized

512 GPU – POD #1 – Rack #1-#18



**Rack #1 - #4, #6-#13, #15-#22, #24-#31, #33-#36 – GPU Racks**  
4 x HPE XD685 M325 GPU Nodes  
Rack Power Estimate: 52 KW

**Rack #5, #23 – L1 BE Network Rack**  
8 x Arista TH5 800G 64p Switch – L1 BE Net  
2 x Arista 64p 100G Switch – FE Net  
1 x Arista 48p 1G Switch – OOB MGMT  
Rack Power Estimate: 20 KW

**Rack #14, #32 – L1 FE Network Rack**  
2 x Arista 64p 100G Switch – FE Net  
1 x Arista 48p 1G Switch – OOB MGMT  
Rack Power Estimate: 2 KW

**Rack #XX – L2 Spine Network Rack**  
8 x Arista TH5 800G 64p Switch – L2 BE Net  
2 x Arista 64p 100G Switch – FE Net Spine  
Rack Power Estimate: 20 KW

**Hardware Power Estimate**  
M325 GPU Node – 13KW  
Arista TH5 800G 64p Switch – 2250W  
Arista 100G 64p Switch – 927W  
Arista 1G 48p Switch – 95W

**Total Solution Power Estimate: 1,704 KW**

512 GPU – POD #2 – Rack #19-#36

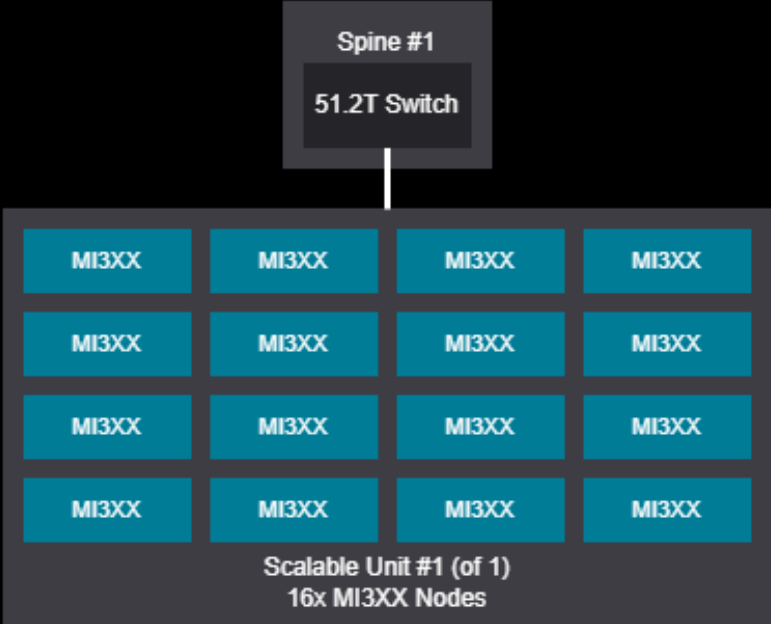


| 1024 GPU M325X Cluster POD |             |             |            |
|----------------------------|-------------|-------------|------------|
| DATE                       | 2023        | BY          | AMD        |
| DESIGNED BY                | AMD         | REVIEWED BY | AMD        |
| APPROVED BY                | AMD         | DATE        | 2023-10-10 |
| PROJECT ID                 | 1024GPU-POD | STATUS      | CONFIRMED  |

# Appendix 2 - Topological Variations

# **128 GPU Generic Topology Design Examples**

# 8-128 GPU Single Switch Design

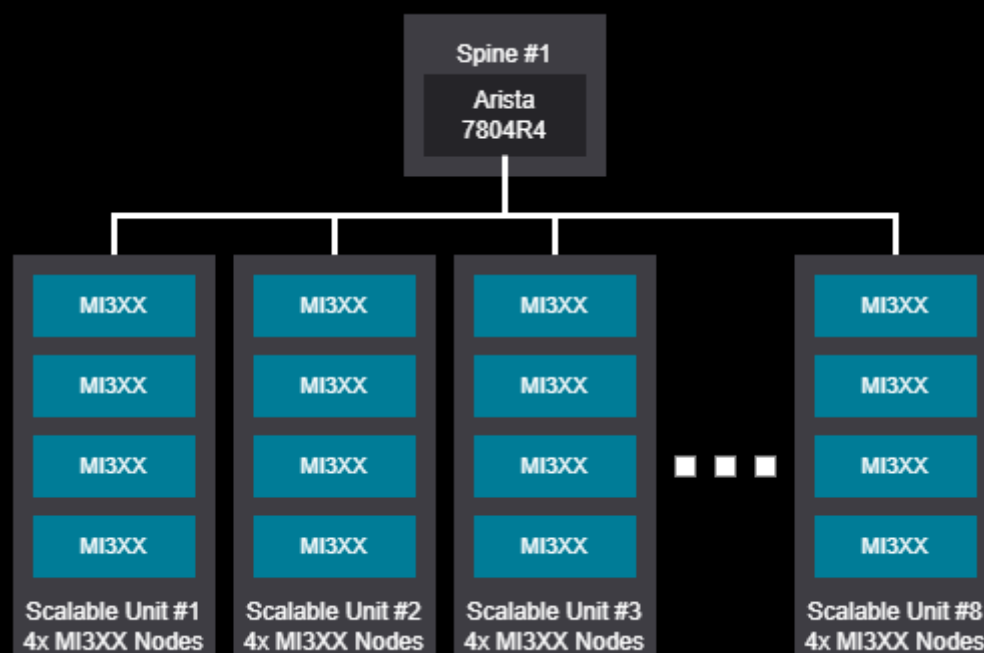


- Can use Cisco G200 or Broadcom Tomahawk 5
- Single switch design will give the best performance possible compared to any alternative
- Extremely simple configuration and deployment

**129-1024 GPU Tree Designs – Scheduled Fabric Specific**

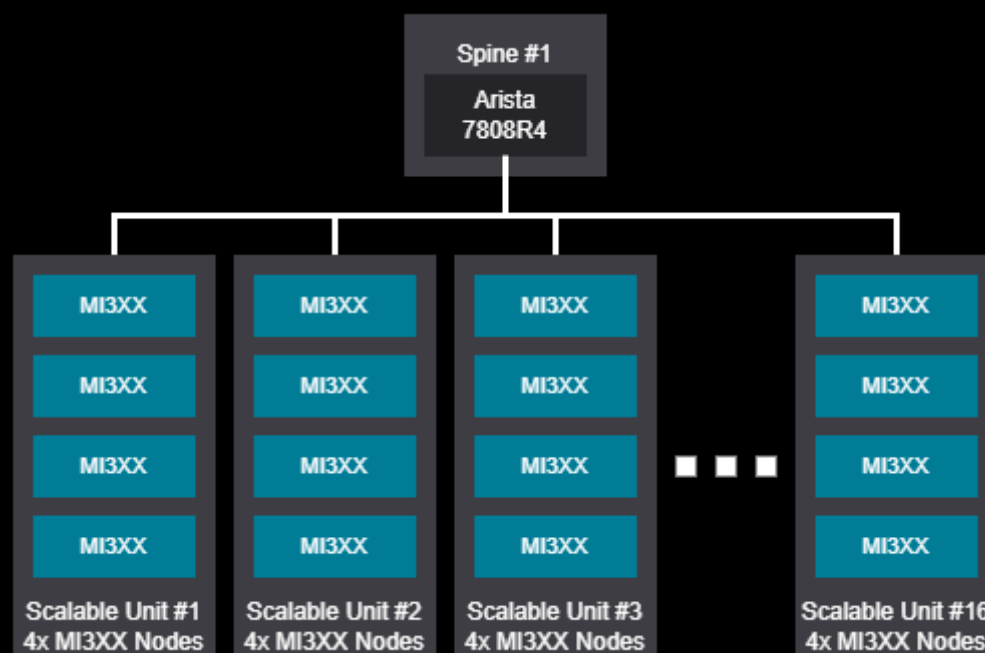


# 129-256 GPU Tree Design



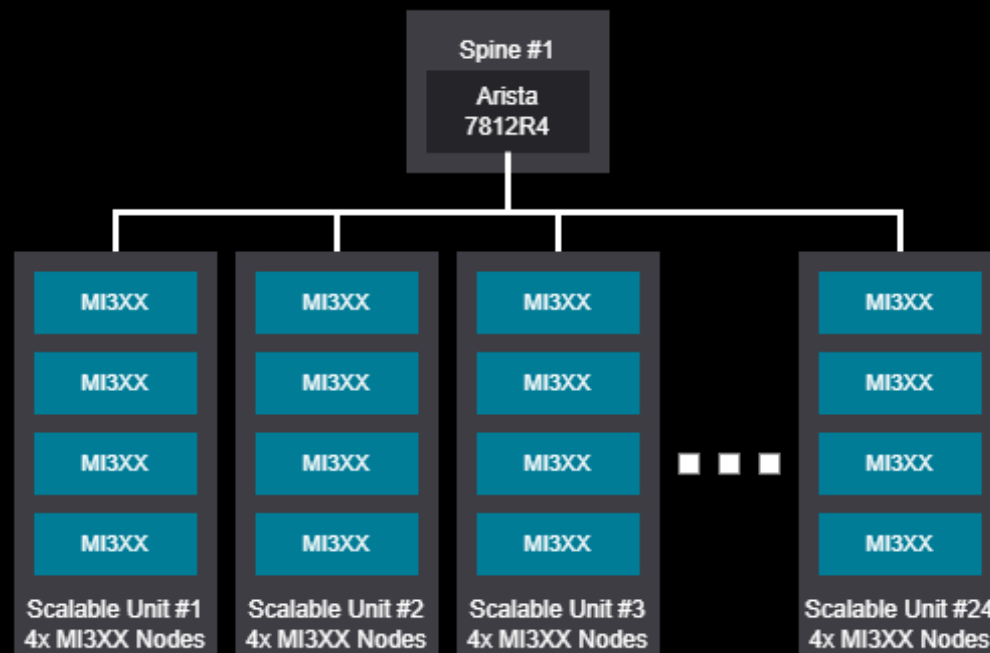
- Built around the Arista 7800R4 Jericho3-AI/Ramon3 series chassis switch
- Functionally a 2-tier network, but managed as a single switch
- Fully scheduled fabric
- Extremely simple configuration and deployment
- Tree vs rail topology is accomplished by adjusting which ports a node is connected to the switch
- Tree can scale to a slightly smaller cluster than rail

# 257-512 GPU Tree Design



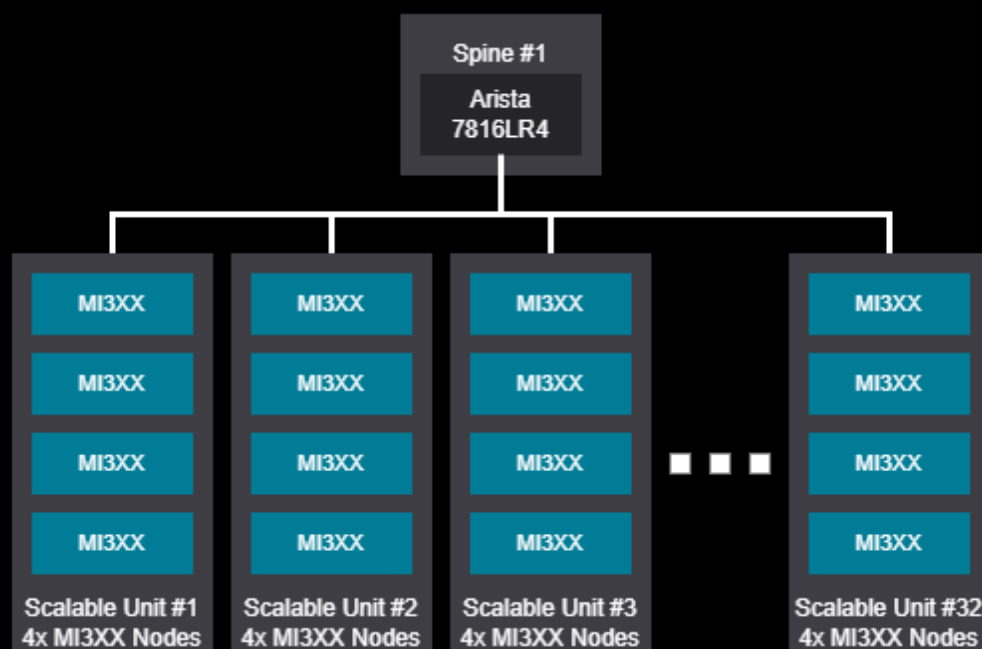
- Built around the Arista 7800R4 Jericho3-AI/Ramon3 series chassis switch
- Functionally a 2-tier network, but managed as a single switch
- Fully scheduled fabric
- Extremely simple configuration and deployment
- Tree vs rail topology is accomplished by adjusting which ports a node is connected to the switch
- Tree can scale to a slightly smaller cluster than rail

# 513-768 GPU Tree Design



- Built around the Arista 7800R4 Jericho3-AI/Ramon3 series chassis switch
- Functionally a 2-tier network, but managed as a single switch
- Fully scheduled fabric
- Extremely simple configuration and deployment
- Tree vs rail topology is accomplished by adjusting which ports a node is connected to the switch
- Tree can scale to a slightly smaller cluster than rail

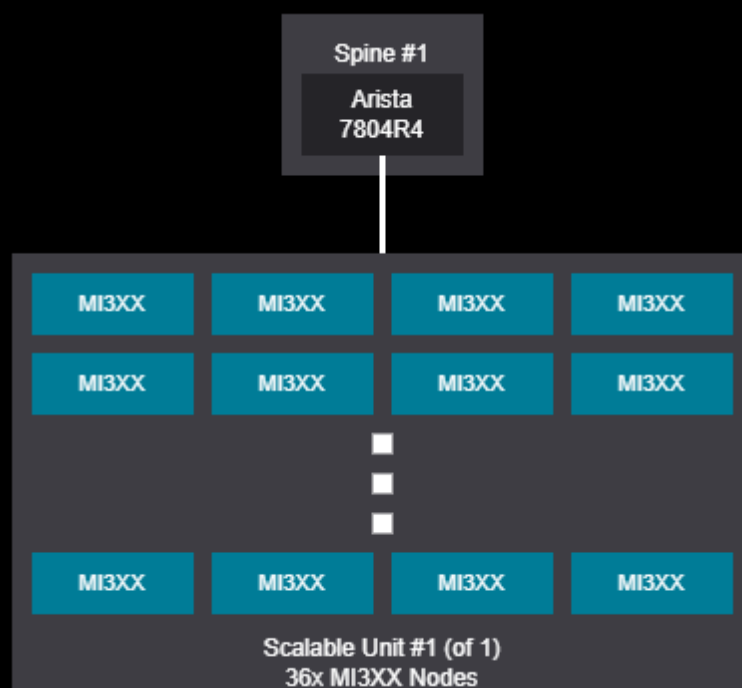
# 769-1024 GPU Tree Design



- Built around the Arista 7800R4 Jericho3-AI/Ramon3 series chassis switch
- Functionally a 2-tier network, but managed as a single switch
- Fully scheduled fabric
- Extremely simple configuration and deployment
- Tree vs rail topology is accomplished by adjusting which ports a node is connected to the switch
- Tree can scale to a slightly smaller cluster than rail

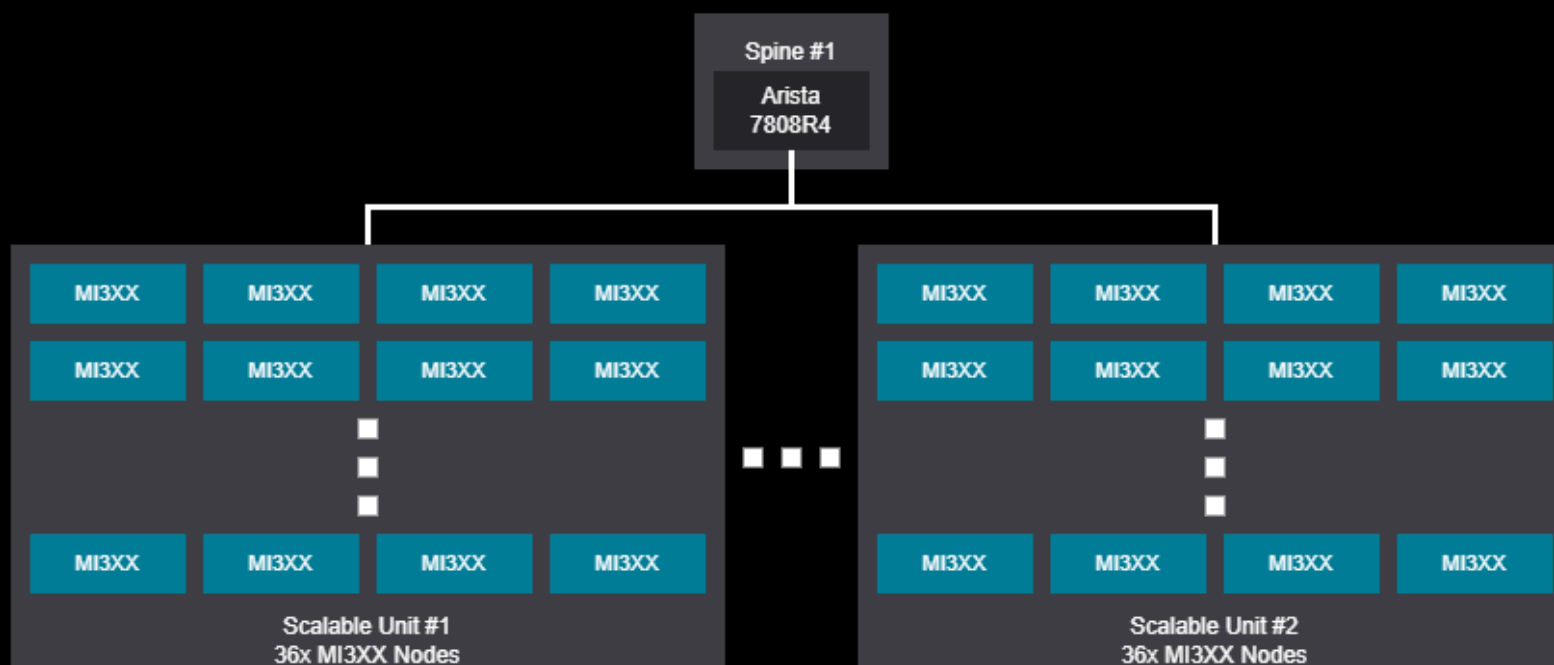
**129-1152 GPU Rail Designs – Scheduled Fabric Specific**

# 129-288 GPU Rail Design



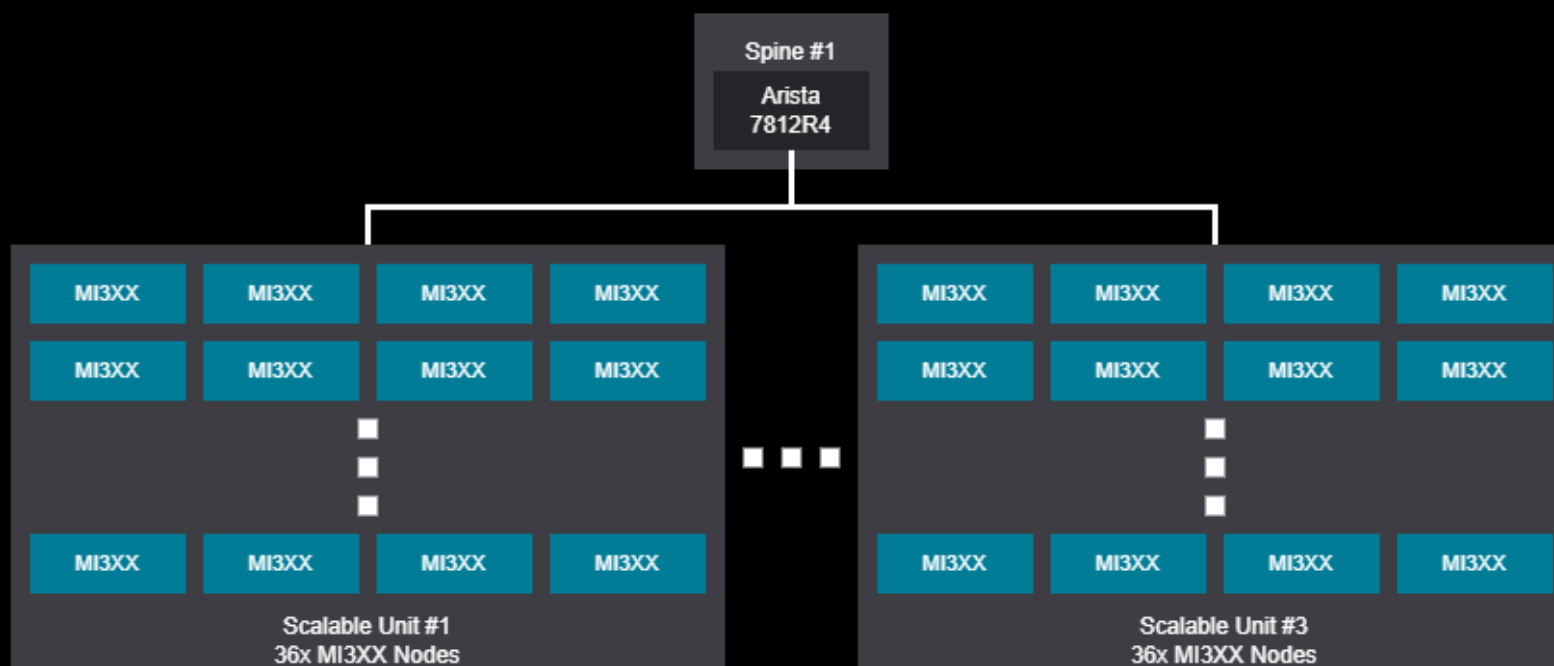
- Built around the Arista 7800R4 Jericho3-AI/Ramon3 series chassis switch
- Functionally a 2-tier network, but managed as a single switch
- Fully scheduled fabric
- Extremely simple configuration and deployment
- Tree vs rail topology is accomplished by adjusting which ports a node is connected to the switch
- Rail can scale to a slightly larger cluster than tree

# 289-576 GPU Rail Design



- Built around the Arista 7800R4 Jericho3-AI/Ramon3 series chassis switch
- Functionally a 2-tier network, but managed as a single switch
- Fully scheduled fabric
- Extremely simple configuration and deployment
- Tree vs rail topology is accomplished by adjusting which ports a node is connected to the switch
- Rail can scale to a slightly larger cluster than tree

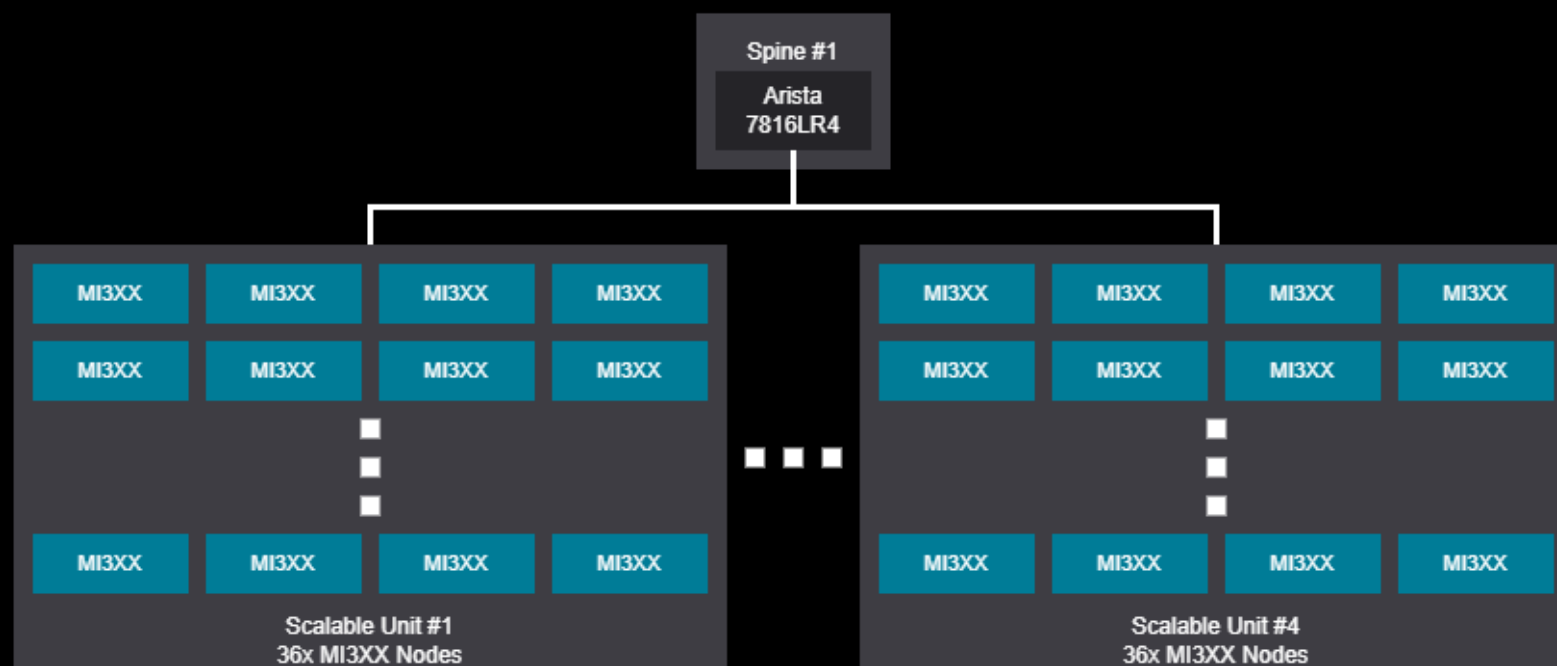
# 577-864 GPU Rail Design



- Built around the Arista 7800R4 Jericho3-AI/Ramon3 series chassis switch
- Functionally a 2-tier network, but managed as a single switch
- Fully scheduled fabric
- Extremely simple configuration and deployment
- Tree vs rail topology is accomplished by adjusting which ports a node is connected to the switch
- Rail can scale to a slightly larger cluster than tree



# 865-1152 GPU Rail Design



- Built around the Arista 7800R4 Jericho3-AI/Ramon3 series chassis switch
- Functionally a 2-tier network, but managed as a single switch
- Fully scheduled fabric
- Extremely simple configuration and deployment
- Tree vs rail topology is accomplished by adjusting which ports a node is connected to the switch
- Rail can scale to a slightly larger cluster than tree

# DISCLAIMER AND ATTRIBUTIONS

## DISCLAIMER

The information contained herein is for informational purposes only, and is subject to change without notice. While every precaution has been taken in the preparation of this document, it may contain technical inaccuracies, omissions and typographical errors, and AMD is under no obligation to update or otherwise correct this information. Advanced Micro Devices, Inc. makes no representations or warranties with respect to the accuracy or completeness of the contents of this document, and assumes no liability of any kind, including the implied warranties of noninfringement, merchantability or fitness for particular purposes, with respect to the operation or use of AMD hardware, software or other products described herein. No license, including implied or arising by estoppel, to any intellectual property rights is granted by this document. Terms and limitations applicable to the purchase or use of AMD's products are as set forth in a signed agreement between the parties or in AMD's Standard Terms and Conditions of Sale. GD-18

## COMPLIANCE WITH LAWS

Customer shall adhere to all applicable export laws and regulations including, without limitation, those administered by the U.S. Department of Commerce – Bureau of Industry and Security (U.S. Export Administration Regulations 15 CFR 730 et seq.) and those administered by the U.S. Department of State in accordance with the U.S. International Traffic in Arms Regulations (ITAR) set forth in Subchapter M, Title 22, Code of Federal Regulations, Parts 120 through 130 (22 CFR 120-130), as the same may be amended from time to time, and shall not export, re-export, resell, transfer, or disclose, directly or indirectly, any Products or technical data, or the direct product of any Products or technical data, to any proscribed person, entity, or country, or foreign national thereof, unless properly authorized by the U.S. government and/or any other applicable or relevant government or regulatory body, including the export authorities of all respective countries. For the avoidance of doubt, Customer shall not use Products in, or re-export Products to Belarus, Russia and the Donetsk (DNR) or Luhansk (LNR) regions of Ukraine, regardless of the applicable export laws and regulations. Customer shall impose upon its customers terms at least as restrictive as those contained in this Clause 14 with respect to any sale, distribution or export of Products.

© 2025 Advanced Micro Devices, Inc. All rights reserved. AMD, the AMD Arrow logo, AMD Instinct, AMD together we advance\_, Infinity Fabric, ROCm, and combinations thereof are trademarks of Advanced Micro Devices, Inc. Amazon S3, Arista, Arista OSFP, APC, Broadcom, Ciena, Cisco, CloudVision, DataDirect Networks, Dell, DriveNets, EOS, FS.com, Hammerspace, Hewlett-Packard Enterprise, IOS, Juniper, JUNOS, Lenovo, Linux, MTP, Netshelter, Nokia, Proliant, Pure Storage, Schneider Electric, SONiC, Super Micro Computer Inc, Tomahawk, Ubuntu, Vast Data, Weka, and other product names used in this publication are for identification purposes only and may be trademarks of their respective owners. Certain AMD technologies may require third-party enablement or activation. Supported features may vary by operating system. Please confirm with the system manufacturer for specific features. No technology or product can be completely secure.